

# MATH 2123 (MATHEMATICS-III)

## 2017 Beginner Guide + Full Step-by-Step Solutions

Khulna University of Engineering & Technology  
Department of Building Engineering and Construction Management  
B.Sc. Engineering 2nd Year 1st Term Regular Examination, 2017

**NOTE:** This guide is written for a beginner. Every main question is organized as Topic -> Basic -> Problem-Solving Procedure -> Worked Solution. Two items in the supplied paper are mathematically inconsistent as printed: Question 6(a) asks for the inverse of a singular matrix, and Question 7(a) contains traffic data that violate total flow conservation. Both are demonstrated carefully rather than guessed.

### How to use this guide

- Read the BASIC box first. It contains the minimum formula or idea needed for the question.
- Memorize the PROCEDURE, not the numerical answer. The same procedure works for similar exam problems.
- Then follow the worked solution line by line and compare the final answer.
- Before the exam, use the Rapid Revision section at the end to recall the method quickly.

### Question map

| Question | Main topic                                             | Key skill                                                           |
|----------|--------------------------------------------------------|---------------------------------------------------------------------|
| 1        | Laplace transform + ODE                                | Kernel, shifting/differentiation, boundary value                    |
| 2        | Inverse Laplace + beam + vector dependence             | Heaviside expansion, unit step, determinant                         |
| 3        | Vector calculus + space curve                          | $\text{grad}(\text{div } A)$ , tangent plane, velocity/acceleration |
| 4        | Vector fields + directional derivative + line integral | curl, unit direction, path integration                              |
| 5        | Matrix forms                                           | matrix definitions, symmetric/skew split, rank                      |
| 6        | Inverse/consistency of linear systems                  | singularity test, Gaussian elimination                              |
| 7        | Traffic flow + eigen-system                            | flow conservation, eigenvalues/eigenvectors                         |
| 8        | Adjoint, Cayley-Hamilton, normal form                  | inverse, characteristic polynomial, PAQ                             |

## Question 1: Integral transform, Laplace transform, and an ODE

| TOPIC                                   | BASIC                                                                                                                                                                                                             | PROCEDURE                                                                                                                                                                                 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Integral transforms and Laplace methods | An integral transform converts $f(t)$ into a new function $F(s)$ by integration with a kernel $K(s,t)$ . For Laplace, $K(s,t)=e^{(-st)}$ . Useful rules: $L\{e^{(-at)}f(t)\}=F(s+a)$ , and $L\{t f(t)\}=-dF/ds$ . | 1) Identify the transform rule. 2) Write the algebraic transform. 3) Simplify in $s$ . 4) For ODEs, transform derivatives, solve for $Y(s)$ , invert, then apply any remaining condition. |

### 1(a) - Integral transform and kernel

A general integral transform of  $f(t)$  may be written as

$$F(s) = \int K(s,t) f(t) dt$$

The function  $K(s,t)$  is called the kernel of the transform. In the Laplace transform,

$$L\{f(t)\} = F(s) = \int_0^{\infty} e^{(-st)} f(t) dt$$

Therefore the Laplace kernel is  $K(s,t)=e^{(-st)}$ .

**FINAL ANSWER:** Kernel of the Laplace transform:  $K(s,t) = e^{(-st)}$ .

### 1(b) - Find $L\{t e^{(-2t)} \sin 3t\}$

First transform  $e^{(-2t)} \sin 3t$ .

$$L\{\sin 3t\} = 3/(s^2+9)$$

$$L\{e^{(-2t)} \sin 3t\} = 3/((s+2)^2+9)$$

Now use  $L\{t f(t)\} = -dF/ds$ .

$$\begin{aligned} L\{t e^{(-2t)} \sin 3t\} &= -d/ds [3/((s+2)^2+9)] \\ &= 6(s+2) / [(s+2)^2+9]^2 \end{aligned}$$

**FINAL ANSWER:**  $L\{t e^{(-2t)} \sin 3t\} = 6(s+2) / [(s+2)^2+9]^2$ .

### 1(c) - Solve $y'' + 9y = \cos 2t$ , $y(0)=1$ , $y(\pi/2)=-1$ by Laplace transform

Because a derivative value at  $t=0$  is not supplied, let  $y'(0)=v$ . We determine  $v$  from  $y(\pi/2)=-1$ .

$$L\{y''\} = s^2 Y - s y(0) - y'(0) = s^2 Y - s - v$$

$$(s^2+9)Y - s - v = s/(s^2+4)$$

$$Y = (s+v)/(s^2+9) + s/[(s^2+4)(s^2+9)]$$

Use partial fractions on the last term:

$$s/[(s^2+4)(s^2+9)] = (1/5)[s/(s^2+4) - s/(s^2+9)]$$

$$\begin{aligned} y(t) &= \cos 3t + (v/3) \sin 3t + (1/5)(\cos 2t - \cos 3t) \\ &= (4/5)\cos 3t + (v/3)\sin 3t + (1/5)\cos 2t \end{aligned}$$

Apply  $t=\pi/2$ :

$$-v/3 - 1/5 = -1 \Rightarrow v/3 = 4/5 \Rightarrow v = 12/5$$

**FINAL ANSWER:**  $y(t) = (4/5)(\cos 3t + \sin 3t) + (1/5)\cos 2t$ .

## Question 2: Inverse Laplace, beam deflection, and linear dependence

| TOPIC                    | BASIC                                                                                                                                                                                                                | PROCEDURE                                                                                                                                                                                                                                                                 |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Three independent skills | Heaviside expansion handles simple linear poles. A beam satisfies $EI y'''' = w(x)$ under the usual sign convention. Three vectors in $\mathbb{R}^3$ are independent if the determinant formed from them is nonzero. | Inverse Laplace: expand $\rightarrow$ invert each standard term. Beam: write load with a unit step $\rightarrow$ transform/integrate $\rightarrow$ impose support conditions. Vectors: place as columns $\rightarrow$ compute det $\rightarrow$ if zero, find a relation. |

### 2(a) - Inverse Laplace using Heaviside expansion

$$F(s) = (11s^2 - 2s + 5)/[(s-2)(2s-1)(s+1)]$$

Assume

$$F(s) = A/(s-2) + B/(2s-1) + C/(s+1)$$

For simple poles, substitute the pole values:

$$\text{At } s=2: A = 45/9 = 5$$

$$\text{At } s=1/2: B = (27/4)/(-9/4) = -3$$

$$\text{At } s=-1: C = 18/9 = 2$$

$$F(s) = 5/(s-2) - 3/(2s-1) + 2/(s+1)$$

Since  $-3/(2s-1) = -(3/2)/(s-1/2)$ ,

$$\text{FINAL ANSWER: } \mathcal{L}^{-1}\{F(s)\} = 5e^{2t} - (3/2)e^{t/2} + 2e^{-t}.$$

### 2(b) - Cantilever beam with load on only the first half

Let  $a=L/2$  and  $q=w_0/(EI)$ . Taking downward deflection as positive, the load can be represented by

$$EI y'''' = w_0[1 - u(x-a)]$$

The clamped end conditions are  $y(0)=0$  and  $y'(0)=0$ . The free end conditions are  $y''(L)=0$  and  $y'''(L)=0$ .

After transforming in  $x$  (or equivalently integrating the unit-step form), write the unknown initial curvature and shear terms as  $C_2=y''(0)$ ,  $C_3=y'''(0)$ :

$$y(x) = C_2 x^2/2 + C_3 x^3/6 + q x^4/24 - q(x-a)^4 u(x-a)/24$$

Apply the free-end conditions at  $x=L$ . For  $x>a$ ,

$$y'''(L) = C_3 + qa = 0 \Rightarrow C_3 = -qa$$

$$y''(L) = C_2 - q a^2/2 = 0 \Rightarrow C_2 = q a^2/2$$

With  $a=L/2$ , the deflection becomes

For  $0 \leq x \leq L/2$ :

$$y(x) = (w_0/EI) [ x^4/24 - L x^3/12 + L^2 x^2/16 ]$$

For  $L/2 \leq x \leq L$ :

$$y(x) = (w_0/EI) [ L^3 x/48 - L^4/384 ]$$

**NOTE:** If your course uses upward-positive deflection, place a minus sign in front of these expressions.

**FINAL ANSWER:** Piecewise deflection as written above; the two pieces have the same value and slope at  $x=L/2$ .

## 2(c) - Linear dependence of two sets of vectors

Set 1:

$$A=(2,1,-3), B=(1,0,-4), C=(4,3,-1)$$

$$\det[A \ B \ C] = 0$$

A nonzero relation is

$$-3A + 2B + C = 0 \Rightarrow C = 3A - 2B$$

Therefore Set 1 is linearly dependent.

Set 2:

$$A=(1,-3,2), B=(2,-4,-1), C=(3,2,-1)$$

$$\det[A \ B \ C] = 41 \neq 0$$

**FINAL ANSWER:** Set 1 is dependent ( $C=3A-2B$ ). Set 2 is linearly independent.

## Question 3: grad(div A), tangent plane, velocity and acceleration

### TOPIC

Vector calculus and space curves

### BASIC

For  $A = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ ,  $\text{div } A = P_x + Q_y + R_z$ .  $\text{grad}(\text{div } A)$  is the gradient of that scalar. A tangent plane to  $F(x,y,z)=0$  has normal  $\text{grad } F$ . Velocity  $\mathbf{r}'(t)$ , acceleration  $\mathbf{r}''(t)$ .

### PROCEDURE

Compute derivatives carefully. For a tangent plane, first move everything to  $F=0$ , compute  $\text{grad } F$  at the point, then use  $\text{grad } F(P) \cdot [(x,y,z)-P] = 0$ . For curves, differentiate componentwise and take magnitudes.

### 3(a) - If $A = \mathbf{r\_vector}/|\mathbf{r}|$ , find $\text{grad}(\text{div } A)$

Let  $r = \sqrt{x^2 + y^2 + z^2}$ . Then

$$A = (x/r)\mathbf{i} + (y/r)\mathbf{j} + (z/r)\mathbf{k}$$

The divergence is

$$\begin{aligned} \text{div } A &= d(x/r)/dx + d(y/r)/dy + d(z/r)/dz \\ &= (1/r - x^2/r^3) + (1/r - y^2/r^3) + (1/r - z^2/r^3) \\ &= 3/r - (x^2 + y^2 + z^2)/r^3 = 2/r \end{aligned}$$

$$\text{grad}(\text{div } A) = \text{grad}(2/r) = -2(x\mathbf{i} + y\mathbf{j} + z\mathbf{k})/r^3$$

**FINAL ANSWER:**  $\text{grad}(\text{div } A) = -2 \mathbf{r\_vector} / r^3$ , for  $r \neq 0$ .

### 3(b) - Tangent plane to $2xz^2 - 3xy - 4x = 7$ at $(1, -1, 2)$

Define

$$F(x,y,z) = 2xz^2 - 3xy - 4x - 7$$

$$F_x = 2z^2 - 3y - 4$$

$$F_y = -3x$$

$$F_z = 4xz$$

At  $(1, -1, 2)$ :

$$\text{grad } F = (7, -3, 8)$$

$$7(x-1) - 3(y+1) + 8(z-2) = 0$$

**FINAL ANSWER:** Tangent plane:  $7x - 3y + 8z = 26$ .

### 3(c) - Magnitude of velocity and acceleration at $t=0$

$$\mathbf{r}(t) = (e^{-t}, 2\cos 3t, 2\sin 3t)$$

$$\mathbf{v}(t) = \mathbf{r}'(t) = (-e^{-t}, -6\sin 3t, 6\cos 3t)$$

$$\mathbf{v}(0) = (-1, 0, 6)$$

$$|\mathbf{v}(0)| = \sqrt{1+36} = \sqrt{37}$$

$$\mathbf{a}(t) = \mathbf{r}''(t) = (e^{-t}, -18\cos 3t, -18\sin 3t)$$

$$\mathbf{a}(0) = (1, -18, 0)$$

$$|a(0)| = \sqrt{1+324} = \sqrt{325} = 5\sqrt{13}$$

**FINAL ANSWER:** Speed at  $t=0$ :  $\sqrt{37}$ . Acceleration magnitude at  $t=0$ :  $5\sqrt{13}$ .

## Question 4: Irrotational field, directional derivative, and line integral

| TOPIC                                | BASIC                                                                                                                                                                                                                             | PROCEDURE                                                                                                                                                                                                                            |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Curl, gradient, and path integration | A vector field is irrotational if $\text{curl } A=0$ .<br>The directional derivative of scalar P along unit vector u is $\text{grad } P \cdot u$ .<br>A line integral is $\int A \cdot dr = \int (P \, dx + Q \, dy + R \, dz)$ . | Curl: compute the three components and test zero. Directional derivative: destination-start $\rightarrow$ normalize $\rightarrow$ dot with gradient. Piecewise path: parameterize each straight segment and add the three integrals. |

### 4(a) - Test whether A is irrotational

$$A = (6xy + z^3)i + (3x^2 - z)j + (3xz^2 - 4)k$$

Let  $P = 6xy + z^3$ ,  $Q = 3x^2 - z$ ,  $R = 3xz^2 - 4$ . Then

$$\begin{aligned} \text{curl } A &= (R_y - Q_z)i + (P_z - R_x)j + (Q_x - P_y)k \\ &= (0 - (-1))i + (3z^2 - 3z^2)j + (6x - 6x)k \\ &= i \end{aligned}$$

**FINAL ANSWER:**  $\text{curl } A = i \neq 0$ . Therefore A is NOT irrotational, so no scalar potential exists.

### 4(b) - Directional derivative of $P = 4e^{(2x-y+z)}$

Starting point  $A = (1, 1, -1)$ , target point  $B = (-3, 5, 6)$ .

$$AB = (-4, 4, 7), \quad |AB| = 9, \quad u = (-4/9, 4/9, 7/9)$$

$$\text{grad } P = 4e^{(2x-y+z)}(2, -1, 1)$$

At  $(1, 1, -1)$ , the exponent is 0:

$$\text{grad } P = (8, -4, 4)$$

$$D_u P = \text{grad } P \cdot u$$

$$= 8(-4/9) + (-4)(4/9) + 4(7/9)$$

$$= -20/9$$

**FINAL ANSWER:** Directional derivative =  $-20/9$ .

### 4(c) - Line integral along three straight segments

$$A = (2y+3)i + xz j + (yz-x)k$$

Segment C1:  $(0, 0, 0) \rightarrow (0, 0, 1)$ . Here  $x=y=0$ ,  $dx=dy=0$ , and  $R=0$ .

$$\text{Integral}_{C1} A \cdot dr = 0$$

Segment C2:  $(0, 0, 1) \rightarrow (0, 1, 1)$ . Here  $x=0, z=1$ ,  $dx=dz=0$ , and  $Q=xz=0$ .

$$\text{Integral}_{C2} A \cdot dr = 0$$

Segment C3:  $(0, 1, 1) \rightarrow (2, 1, 1)$ . Here  $y=z=1$ ,  $dy=dz=0$ .

$$A \cdot dr = (2y+3)dx = 5 \, dx$$



$$\text{Integral\_C3} = \int_0^5 dx = 10$$

**FINAL ANSWER:**  $\text{Integral\_C A.dr} = 10.$

## Question 5: Matrix definitions, symmetric/skew decomposition, and rank

| TOPIC                          | BASIC                                                                                                                                                                                                     | PROCEDURE                                                                                                                                                                                                                          |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Matrix structure and reduction | <p>A symmetric matrix satisfies <math>A^T=A</math>; skew-symmetric satisfies <math>A^T=-A</math>. Every square matrix splits as <math>(A+A^T)/2 + (A-A^T)/2</math>. Rank equals the number of pivots.</p> | <p>For decomposition use the two standard formulas. For rank, use elementary row operations to echelon form. Continue to RREF for canonical form, then use column operations to turn the remaining nonpivot columns into zero.</p> |

### 5(a) - Definitions with examples

1) Skew-symmetric matrix:  $A^T=-A$ ; diagonal entries must be zero.

Example:  $\begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$

2) Principal submatrix: obtained by selecting the same index set of rows and columns.

Example: from  $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ , rows/cols  $\{1,3\}$  give  $\begin{bmatrix} 1 & 3 \\ 7 & 9 \end{bmatrix}$ .

3) Elementary matrix: obtained by performing one elementary row operation on an identity matrix.

Example:  $\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$  represents  $R_2 \rightarrow R_2 + 3R_1$ .

4) Diagonal matrix: all off-diagonal entries are zero.

Example:  $\text{diag}(2, -1, 5)$ .

### 5(b) - Write $A=P+Q$ with $P$ symmetric and $Q$ skew-symmetric

$$A = \begin{bmatrix} 3 & 3 & -1 \\ 0 & 3 & -2 \\ -1 & 2 & 2 \end{bmatrix}$$

$$P = (A+A^T)/2$$

$$Q = (A-A^T)/2$$

$$P = \begin{bmatrix} 3 & 3/2 & -1 \\ 3/2 & 3 & 0 \\ -1 & 0 & 2 \end{bmatrix}$$

$$Q = \begin{bmatrix} 0 & 3/2 & 0 \\ -3/2 & 0 & -2 \\ 0 & 2 & 0 \end{bmatrix}$$

**FINAL ANSWER:**  $A=P+Q$  with  $P$  and  $Q$  exactly as above.

### 5(c) - Echelon, canonical and normal forms; find rank

$$A = \begin{bmatrix} 2 & 1 & 5 & 1 & 5 \\ 1 & 1 & -3 & -4 & -1 \\ 3 & 6 & -2 & 1 & 8 \\ 2 & 2 & 2 & -3 & 2 \end{bmatrix}$$

A convenient reduction:

- $R_1 \leftrightarrow R_2$ .
- $R_2 \rightarrow R_2 - 2R_1$ ,  $R_3 \rightarrow R_3 - 3R_1$ ,  $R_4 \rightarrow R_4 - 2R_1$ .

3.  $R_2 \rightarrow -R_2$ .
4.  $R_3 \rightarrow R_3 - 3R_2$ .
5. Swap the new  $R_3$  and  $R_4$ .
6.  $R_4 \rightarrow R_4 - 5R_3$ .

One echelon form is

$$\begin{bmatrix} 1 & 1 & -3 & -4 & -1 \\ 0 & 1 & -11 & -9 & -7 \\ 0 & 0 & 8 & 5 & 4 \\ 0 & 0 & 0 & 15 & 12 \end{bmatrix}$$

There are four nonzero rows, so  $\text{rank}(A)=4$ .

Continuing Gauss-Jordan elimination gives the row-canonical (RREF) form

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 1/5 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 4/5 \end{bmatrix}$$

Finally use the column operation  $C_5 \rightarrow C_5 - 2C_1 - (1/5)C_2 - (4/5)C_4$ .

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

**FINAL ANSWER:**  $\text{rank}(A)=4$ ; normal form is  $[I_4 \mid 0]$ .

## Question 6: Inverse of a matrix and consistency of a linear system

### TOPIC

Singularity and Gaussian elimination

### BASIC

A square matrix has an inverse only when  $\det(A) \neq 0$ , equivalently  $\text{rank}(A)=n$ . A linear system is consistent if  $\text{rank}(A)=\text{rank}([A|b])$ . If this common rank equals the number of unknowns, the solution is unique.

### PROCEDURE

Before spending time finding an inverse, check for obvious dependent rows. For a system, row-reduce the augmented matrix and inspect pivots/contradictions.

### 6(a) - Find inverse by elementary transformation

$$A = \begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -3 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix}$$

**NOTE:** As printed, this matrix is singular even though the question asks for its inverse. This is a useful exam check: always test invertibility before doing a long inverse calculation.

Observe the rows:

$$R_2 + R_4 = [1, 1, -1, -3] + [-1, 1, 2, 6] = [0, 2, 1, 3] = R_1$$

So the rows are linearly dependent. Therefore  $\text{rank}(A) < 4$  and  $\det(A) = 0$ .

**FINAL ANSWER:**  $A^{-1}$  does not exist for the matrix printed in the paper.

### 6(b) - Consistency and solution of the system

System:

$$4x - 5y + 6z = 8$$

$$x + y - 3z = -1$$

$$15x - 3y + 9z = 21$$

Common methods for a nonhomogeneous system include substitution/elimination, Gaussian elimination, Gauss-Jordan elimination, inverse-matrix method (if invertible), and Cramer's rule (square nonsingular case).

Use Gaussian elimination on the augmented matrix. Swap the first two equations:

$$[1, 1, -3 \mid -1]$$

$$[4, -5, 6 \mid 8]$$

$$[15, -3, 9 \mid 21]$$

Eliminate x:

$$R_2 \rightarrow R_2 - 4R_1 = [0, -9, 18 \mid 12]$$

$$R_3 \rightarrow R_3 - 15R_1 = [0, -18, 54 \mid 36]$$

Then eliminate y:

$$R3 \rightarrow R3 - 2R2 = [0, 0, 18 \mid 12]$$

$$18z = 12 \Rightarrow z = 2/3$$

$$-9y + 18(2/3) = 12 \Rightarrow y = 0$$

$$x + y - 3z = -1 \Rightarrow x = 1$$

**FINAL ANSWER:** The system is consistent with a unique solution:  $(x, y, z) = (1, 0, 2/3)$ .

## Question 7: Traffic flow and eigenvalues/eigenvectors

| TOPIC                                   | BASIC                                                                                                                                                 | PROCEDURE                                                                                                                                                                      |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conservation equations and eigensystems | At each traffic junction: total inflow = total outflow. Eigenvalues satisfy $\det(A - \lambda I) = 0$ ; eigenvectors satisfy $(A - \lambda I)v = 0$ . | Traffic: write one balance equation per node and check total external inflow/outflow first. Eigen: find characteristic roots, then solve one homogeneous system for each root. |

### 7(a) - Traffic flow in the one-way network

Using the arrows in the supplied diagram, node balances are:

$$\text{Top-left: } 20 + x_2 = 30 + x_1 \Rightarrow x_2 - x_1 = 10$$

$$\text{Top-right: } x_1 + x_4 = 25 + 45 \Rightarrow x_1 + x_4 = 70$$

$$\text{Bottom-left: } 35 + x_3 = 39 + x_2 \Rightarrow x_3 - x_2 = 4$$

$$\text{Bottom-right: } 50 + 40 = x_3 + x_4 \Rightarrow x_3 + x_4 = 90$$

From the first three equations:

$$x_2 = x_1 + 10$$

$$x_3 = x_2 + 4 = x_1 + 14$$

$$x_4 = 70 - x_1$$

Therefore

$$x_3 + x_4 = (x_1 + 14) + (70 - x_1) = 84$$

But the bottom-right balance requires  $x_3 + x_4 = 90$ . This produces the contradiction  $84 = 90$ .

The same issue is visible globally: the printed external inflow is  $20 + 35 + 40 + 50 = 145$ , while the printed external outflow is  $30 + 25 + 45 + 39 = 139$ .

**NOTE:** Hence one of the traffic numbers/arrows in the printed question is almost certainly a typo. No numerical value should be invented without the corrected diagram.

**FINAL ANSWER:** As printed, the traffic system is inconsistent: there is no steady traffic-flow solution, so it is not a uniqueness case.

### 7(b) - Eigenvalues and associated eigenvectors

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{bmatrix}$$

Characteristic polynomial:

$$\begin{aligned} \det(A - \lambda I) &= -\lambda [(1 - \lambda)(4 - \lambda) - 4] \\ &= -\lambda [\lambda^2 - 5\lambda] \\ &= -\lambda^2(\lambda - 5) \end{aligned}$$

Thus  $\lambda = 0$  has algebraic multiplicity 2, and  $\lambda = 5$  has multiplicity 1.

For  $\lambda=0$ :

$Av=0$  gives  $x-2z=0$ , so  $x=2z$  and  $y$  is free.

$$E_0 = \text{span}\{(2,0,1), (0,1,0)\}$$

For  $\lambda=5$ :

$(A-5I)v=0$  gives  $y=0$  and  $2x+z=0$ .

Choose  $z=2 \Rightarrow v=(-1,0,2)$ .

**FINAL ANSWER:** Eigenvalues: 0,0,5. Eigenspace for 0:  $\text{span}\{(2,0,1), (0,1,0)\}$ . An eigenvector for 5 is  $(-1,0,2)$ .

## Question 8: Adjoint/inverse, Cayley-Hamilton theorem, and normal form

| TOPIC              | BASIC                                                                                                                                                                                                                                                                 | PROCEDURE                                                                                                                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Three matrix tools | $A^{-1} = \text{adj}(A)/\det(A)$ when $\det(A) \neq 0$ .<br>Cayley-Hamilton says every square matrix satisfies its own characteristic equation. For an $m \times n$ matrix of rank $r$ , row/column operations can reduce it to the normal form with $I_r$ and zeros. | Adjoint: cofactors $\rightarrow$ transpose. C-H: characteristic polynomial $\rightarrow$ substitute $A$ . Normal form: row-reduce while tracking the row-operation matrix $P$ ; choose $Q$ to scale/finish columns. |

### 8(a) - Find $\text{adj } A$ and hence $A^{-1}$

$$A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

The determinant is

$$\det(A) = -2$$

The adjoint is

$$\text{adj}(A) = \begin{bmatrix} -1 & 1 & -1 \\ 0 & 0 & -2 \\ -1 & -1 & 1 \end{bmatrix}$$

$$\begin{aligned} A^{-1} &= \text{adj}(A)/(-2) \\ &= \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & -1/2 \end{bmatrix} \end{aligned}$$

**FINAL ANSWER:**  $A^{-1} = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & -1/2 \end{bmatrix}$ .

### 8(b) - Verify the Cayley-Hamilton theorem

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

Characteristic equation:

$$\begin{aligned} \det(\lambda I - A) &= \begin{vmatrix} \lambda - 1 & -1 \\ -1 & \lambda - 2 \end{vmatrix} \\ &= (\lambda - 1)(\lambda - 2) - 1 \\ &= \lambda^2 - 3\lambda + 1 \end{aligned}$$

Cayley-Hamilton predicts

$$\begin{aligned} A^2 - 3A + I &= 0 \\ A^2 &= \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} \\ A^2 - 3A + I &= \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} - \begin{bmatrix} 3 & 3 \\ 3 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

**FINAL ANSWER:** Verified:  $A^2 - 3A + I = 0$ .



### 8(c) - Find nonsingular P and Q so that PAQ is normal form

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{bmatrix}$$

The matrix has rank 3, so its 4x3 normal form is

$$N = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

One valid choice is

$$P = \begin{bmatrix} 1 & 5 & -4 & 0 \\ -5 & -1 & 8 & 0 \\ 7 & -1 & -4 & 0 \\ -18 & -6 & 12 & 12 \end{bmatrix}$$

$$Q = (1/12) I_3$$

These are nonsingular because  $\det(P)=1728$  and  $\det(Q)=1/1728$ .

$$P A Q = N$$

**FINAL ANSWER:** One valid pair is the P above and  $Q=(1/12)I_3$ , giving  $PAQ=[I_3;0]$ .

## Rapid Revision: procedures to memorize

| Topic                  | Procedure                                                                                                   |
|------------------------|-------------------------------------------------------------------------------------------------------------|
| Laplace kernel         | $L\{f\} = \int_0^\infty e^{-st} f(t) dt$ ; kernel $= e^{-st}$ .                                             |
| Multiplication by t    | If $L\{f\} = F$ , then $L\{t f\} = -F'(s)$ .                                                                |
| ODE by Laplace         | Transform derivatives $\rightarrow$ solve $Y(s) \rightarrow$ inverse $\rightarrow$ use remaining condition. |
| Heaviside expansion    | For simple poles, coefficient at a pole is obtained by cancel-and-substitute.                               |
| Beam with step load    | Write $w(x)$ using $u(x-a)$ ; impose support/free-end conditions after integration or inversion.            |
| Linear independence    | For three $\mathbb{R}^3$ vectors, $\det \neq 0$ means independent.                                          |
| Tangent plane          | $F=0 \rightarrow \text{grad } F \text{ at point} \rightarrow \text{grad } F \cdot (r-r_0)=0$ .              |
| Directional derivative | $u = (\text{target-start})/ \text{target-start} $ ; answer $= \text{grad } f \cdot u$ .                     |
| Irrotational field     | Compute curl; scalar potential exists only if $\text{curl}=0$ on a suitable domain.                         |
| Matrix rank            | Row reduce; number of pivots/nonzero echelon rows = rank.                                                   |
| System consistency     | Compare $\text{rank}(A)$ and $\text{rank}([A b])$ ; unique if common rank = number of unknowns.             |
| Eigenvalues            | $\det(A - \lambda I) = 0$ , then solve $(A - \lambda I)v = 0$ .                                             |
| Cayley-Hamilton        | Find characteristic polynomial $p(\lambda)$ , then verify $p(A) = 0$ .                                      |
| Normal form            | Rank $r$ matrix $\rightarrow$ by row/column operations $\rightarrow$ block with $I_r$ and zeros.            |

## Answers at a glance

- 1(a) Laplace kernel  $e^{-st}$ .
- 1(b)  $6(s+2)/[(s+2)^2+9]^2$ .
- 1(c)  $y = (4/5)(\cos 3t + \sin 3t) + (1/5)\cos 2t$ .
- 2(a)  $5e^{2t} - (3/2)e^{t/2} + 2e^{-t}$ .
- 2(b)  $y = (w_0/EI)[x^4/24 - Lx^3/12 + L^2x^2/16]$  for  $x \leq L/2$ ;  $y = (w_0/EI)[L^3x/48 - L^4/384]$  for  $x \geq L/2$ .
- 2(c) First set dependent:  $C = 3A - 2B$ . Second set independent.
- 3(a)  $\text{grad}(\text{div } A) = -2 \mathbf{r}_{\text{vector}}/r^3$ .
- 3(b)  $7x - 3y + 8z = 26$ .
- 3(c)  $|v(0)| = \sqrt{37}$ ,  $|a(0)| = 5\sqrt{13}$ .
- 4(a)  $\text{curl } A = \mathbf{i}$ , so not irrotational.
- 4(b)  $-20/9$ .
- 4(c) 10.
- 5(b)  $P = (A + A^T)/2$ ,  $Q = (A - A^T)/2$  as listed.
- 5(c) rank=4; normal form  $[I_4|0]$ .
- 6(a) No inverse as printed because  $R_1 = R_2 + R_4$ .
- 6(b)  $(x, y, z) = (1, 0, 2/3)$ .
- 7(a) No steady solution as printed; traffic data are inconsistent.
- 7(b) eigenvalues 0, 0, 5;  $E_0 = \text{span}\{(2, 0, 1), (0, 1, 0)\}$ ,  $E_5 = \text{span}\{(-1, 0, 2)\}$ .
- 8(a)  $A^{-1} = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & -1/2 \end{bmatrix}$ .
- 8(b)  $A^2 - 3A + I = 0$ .
- 8(c) P and Q listed in the worked solution give  $PAQ = [I_3|0]$ .

## Exam strategy for a beginner

- Start with definitions and direct formula questions; they give reliable marks quickly.
- In matrix questions, always check determinant/rank before attempting an inverse.

- In vector calculus, write  $P$ ,  $Q$ ,  $R$  separately before computing div or curl.
- For directional derivatives, never forget to normalize the direction vector.
- For Laplace problems, keep a small standard-transform table in memory and use shifting rules.
- For a long reduction, write every row operation. Even if arithmetic later goes wrong, method marks remain visible.